

More Power To You!

Have you bothered about what power supply your dealer bundled with your computer? If you have, skip the following! But if you don't even know what brand of PS you have, read on...

To begin with, I just love the Mumbai monsoon: it allows me to take a break from my usual routine and sit by the window pane sipping coffee. But somehow, my computer back home doesn't like the rains—it breaks down every year when the rains come!

Having stacked my cabinet with three hard drives, a deluxe motherboard, two DVD-Writers and a GeForce 7900 card, my not-so-well-powered supply died when it tried to satiate the demand. I still blame it on the rains! Now, I can live without food or water, but without a computer is out of the question.

While there is no dearth of options as far as the availability of brands or models is concerned, the real problem is finding a good, reliable power supply. In the absence of a proper test from our Test Centre, or for that matter by any technology publication in India, spurious, low-quality power supplies (PSes) are handed across the counter and are well-received. With hardware getting power-hungry, these low quality PSes generally fail to deliver when overloaded.

Also, over the past year, the hardware industry has undergone a remarkable change. It has seen the launch of a myriad new chipsets (the 945, 975, and more), new technologies in graphics cards (SLI and CrossFire), new processor architectures (dual-core), and so forth. The resulting new components require a power supply capable of providing sustainable, clean power—since the system as a whole requires more power than it used to earlier.

While a 350W power supply was *de facto* a year ago, and a 400W sold for a premium, with the new hardware requiring more power, these PSes are being replaced by 450W, 500W and 600W beasts. The first question that springs up is, what wattage do I need? If you're building a regular PC with onboard graphics, a single hard drive and an optical drive, then a 400 W PS is more than enough to supply your demand, even at full load. On the other hand, if your PC has all the latest and best hardware such as SLI graphics cards and a top-end processor, make sure your system has at least a 500W power supply.

A higher wattage rating, of course, does not necessarily mean a better PS, so this shouldn't be used as a rule of thumb when buying one. What's more important is the efficiency of the PS. As of today, most high-end PSes available abroad typically have efficiencies in the 70 per cent bracket. Some new PSes have been able to breach the 80 per cent barrier. What does that mean? Well, if your power supply draws 100 watts of energy from the mains, it will deliver only 70

watts to the components inside; the remaining 30 per cent is wasted as heat. While the 70 per cent is for PSes available abroad, the PSes in the Indian market are not a patch on them in terms of efficiency. There are certain brands that we know are good for sure, and we shall mention these later.

There are quite a few things that have cropped up in the power supply category. To reduce wire clutter inside the cabinet, manufacturers have come up with detachable wires. The wires have connectors on both ends, where one end plugs into the PS and the other into the respective devices. Gamers can soup up their cabinets by putting up PSes with neon lights on them. Most importantly, the new PSes support ATX v2.0. All new motherboards require 24-pin power connectors and not the 20-pin ones used by older motherboards.

Coming down to market realities, Mercury, HIS, PERX, Frontech, and iBall sell cabinets that ship with PSes, unless you are smart enough to opt for something better. In our office of 200 people, we have used all these brands and have found nothing wrong with them: they work for three to four years without problems—unless one stresses them with loads of hardware. But from our test lab experience, we can definitely vouch for Antec for their quality of PSes and performance under load. Currently, this is the only international brand available in the Indian market, and is expensive. However, considering the fact that one doesn't upgrade a power supply every six months, spending some extra cash is money well spent.

VIP makes good power supplies, and we have a couple of them running our test benches 24x7. I'd recommend VIP to those who can't afford Antec but want a reliable power supply. Zebronic, an Indian company, also offers good power supplies, especially their Diamond series. Here, I'm listing some good brands and models, with prices in rupees. ■



Illustration Pradip Ingale

Power Supplies: Brands & Prices				
Brand	400W	500W	600W	720W
Frontech	Rs.600	800	1,200	NA
Zebronic	Rs.1,500	1,900	5,500	7,000
VIP	Rs.1,500	2,800	5,000	NA
Antec Models				Price (Rs.)
Antec Phantom 350				10,717
Antec Smart Power SP 350				2,854
Antec Smart Power SP 400				3,636
Antec True Control 550				8,037
Antec True Power TP 550				7,576

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